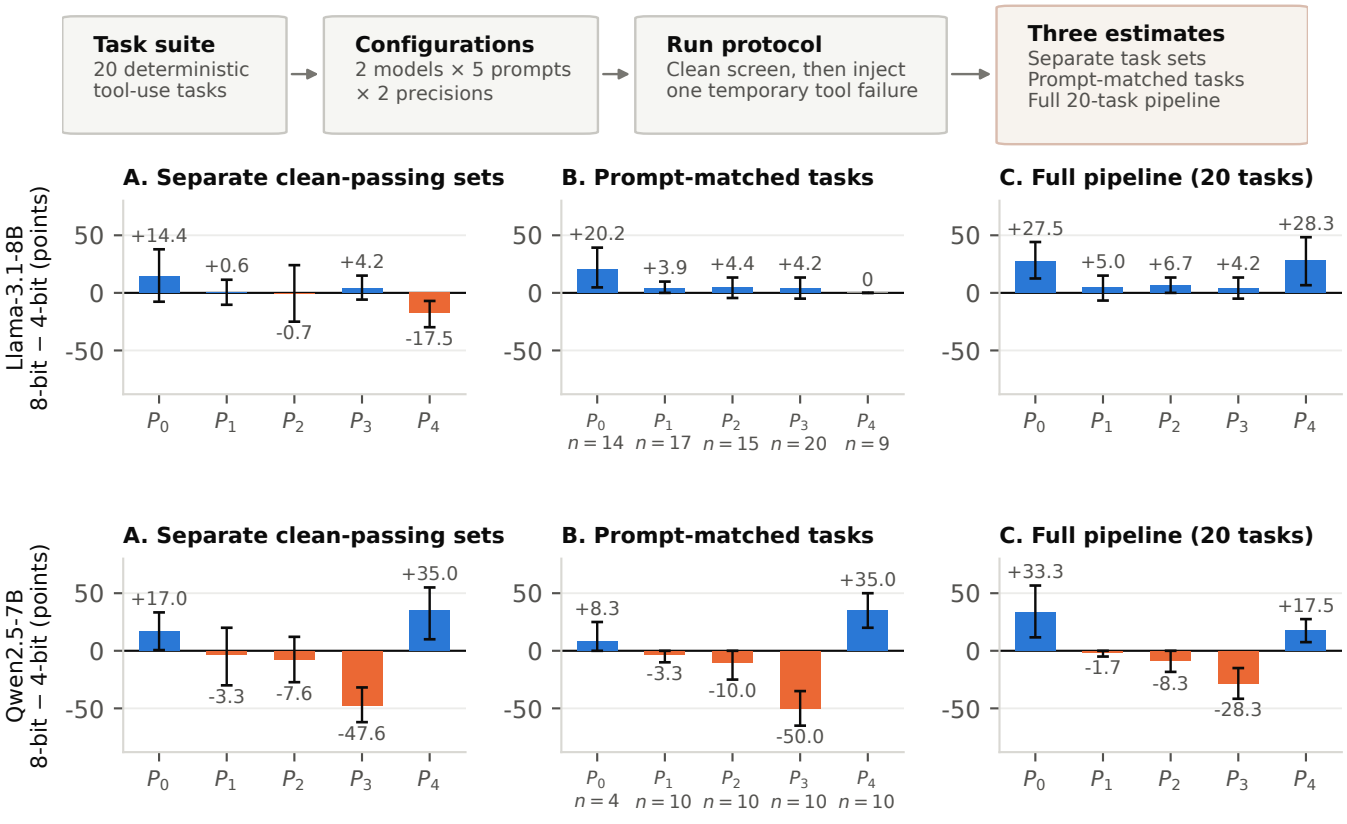
Experiment and evaluation targets
Task suite
20 deterministic tool-use tasks
Configurations
2 models × 5 prompts × 2 precisions
Run protocol
Clean screen, then inject one temporary tool failure
Three estimates
Separate task sets
Prompt-matched tasks
Full 20-task pipeline
Llama-3.1-8B
8-bit − 4-bit (points)
A. Separate clean-passing sets
+14.4
+0.6
-0.7
+4.2
-17.5
B. Prompt-matched tasks
+20.2
+3.9
+4.4
+4.2
0
n = 14
n = 17
n = 15
n = 20
n = 9
C. Full pipeline (20 tasks)
+27.5
+5.0
+6.7
+4.2
+28.3
Qwen2.5-7B
8-bit − 4-bit (points)
A. Separate clean-passing sets
+17.0
-3.3
-7.6
-47.6
+35.0
B. Prompt-matched tasks
+8.3
-3.3
-10.0
-50.0
+35.0
n = 4
n = 10
n = 10
n = 10
n = 10
C. Full pipeline (20 tasks)
+33.3
-1.7
-8.3
-28.3
+17.5
50
0
-50
P0
P1
P2
P3
P4